

Arnav Magadala

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Education

California Polytechnic State University (Cal Poly SLO)

B.S Mechanical Engineering, GPA: 3.35

San Luis Obispo, CA

Graduation: May 2028

Experience

Prototype Vehicles Laboratory (PROVE)

Automotive Engineer

Jan 2025 – Present

San Luis Obispo, CA

- Designed and modeled custom wheel spacers in SolidWorks based on motor dimensional and fitment requirements
- Prototyped wheel spacers through additive manufacturing; fabricated aluminum component using waterjet cutting
- Developed a temporary inverter mounting system in SolidWorks to support vehicle integration and testing activities
- Fabricated and installed a temporary battery enclosure lid and inverter mounting assembly
- Designed and integrated a front crash structure utilizing honeycomb aluminum to improve impact protection
- Researched thermal requirements and developed a cooling system test procedure for EV powertrain evaluation

Cal Poly Space Systems (CPSS)

Aerospace Engineer- Center Stage Team Lead

Jan 2026 – Present

San Luis Obispo, CA

- Designed and modeled aerodynamic control surfaces (wing canards and vertical stabilizers) using SolidWorks
- Developed custom airfoil profiles, translating 3-D geometry to manufacturable 2-D toolpaths
- Generated and validated foam-cutting toolpaths using DevFoam and Mach3 software to manufacture wing structures
- Designed and manufactured custom manual hot wire for rapid prototyping of delta-wing configurations
- Designed and 3D printed lightweight camera fairing to mount and protect onboard imaging systems during flight
- Assisted in booster rocket assembly, performing composite fiberglass layups, finishing, and structural integration
- Researching, designing, and testing booster separation system; managing a team and overseeing multiple projects
- Conducted post-flight failure analysis through video review, hardware inspection, and root-cause investigation

Cal Poly Racing- Formula SAE

Mechanical Design Engineer

Sep 2024 – Feb 2025

San Luis Obispo, CA

- Redesigned the brake pedal assembly in Siemens NX to improve driver ergonomics
- Optimized throttle pedal geometry in Siemens NX, reducing throttle pedal mass by 15%

ItoI Data

Project Researcher and Coder

Aug 2023 – Sep 2023

Moorpark, CA

- Identified opportunities for AI-driven solutions by analyzing data accessibility challenges faced by local businesses.
- Created a website that scrapes online data to quickly consolidate online company information

Pilot Plant

Student Worker

April 2026 – Present

San Luis Obispo, CA

- Operated and maintained food processing equipment including flow wrappers, tempering systems, etc.
- Performed ATP and pH testing to verify sanitation compliance and food safety standards
- Utilized precision measurement equipment to accurately weigh ingredients and support production operations

Volunteering/ Extracurriculars

Cal Poly Indian Students Association |Logistics Lead

Sep 2025 – Present

- Coordinate with university administration to secure venues, permits, and event approvals
- Create and manage shift schedules to ensure smooth event operations
- Lead vendor outreach for cultural events, also promoting events on social media

African Library Project (ALP) |Project Lead Manager

June 2020 – Present

- Led a team of 20+ volunteers, coordinating logistics and outreach
- Collected 3,000+ books for a new library in Lesotho

Skills

- Software:** Siemens-NX, SolidWorks, Fusion 360, Dev Foam, Mach3, Microsoft Excel
- Programming/Analysis:** Python, MATLAB, Finite Element Analysis (FEA)
- Manufacturing:** Lathe, Mill, Welding, Fiberglass layups, Waterjet cutting, Wood working, 3-D printing
- Engineering:** Mechanical Design, CAD, Prototyping, Failure Analysis, Design for Manufacturing (DFM)

Awards

- Sigma Phi Epsilon-Balanced Man:** Recognized for leadership, academic excellence, and personal development
- Cal Poly Leadership Award:** Recognized for completing Emerging Series course and school community service